

AP Precalculus

Unit 3: Trigonometric and Polar Functions

Lesson 3.7 Sinusoidal Function Context and Data Modeling

Name: _____ Date: _____

Period: _____

Learning Targets — I Can...

1. I can determine the amplitude, vertical shift (midline), period, and phase shift of a sinusoidal model from a data table using the formulas $A = \frac{\max - \min}{2}$ and $D = \frac{\max + \min}{2}$. (LO 3.7.A)
2. I can write a sinusoidal function model $f(x) = A \sin(B(x - C)) + D$ (or cosine form) that fits a set of periodic real-world data. (LO 3.7.A)
3. I can use a sinusoidal model to predict output values within the contextual domain and evaluate whether my predictions are reasonable. (LO 3.7.A)

What's in This Packet

#	Component	Description	Score
1	Warm-Up	Spiral review: reading sinusoidal graphs; computing amplitude and midline from data	_____
2	Guided Notes	Extracting parameters from data; writing and validating a sinusoidal model	_____
3	Group Activity	Build and evaluate sinusoidal models from real-world data (tiered R/A/E)	_____
4	Exit Ticket	Identify amplitude, period, and midline from a small data table	_____
5	Homework	Mixed: parameter extraction, prediction, AP-style MC	_____
Total			_____